Mechatronics Design Laboratory I

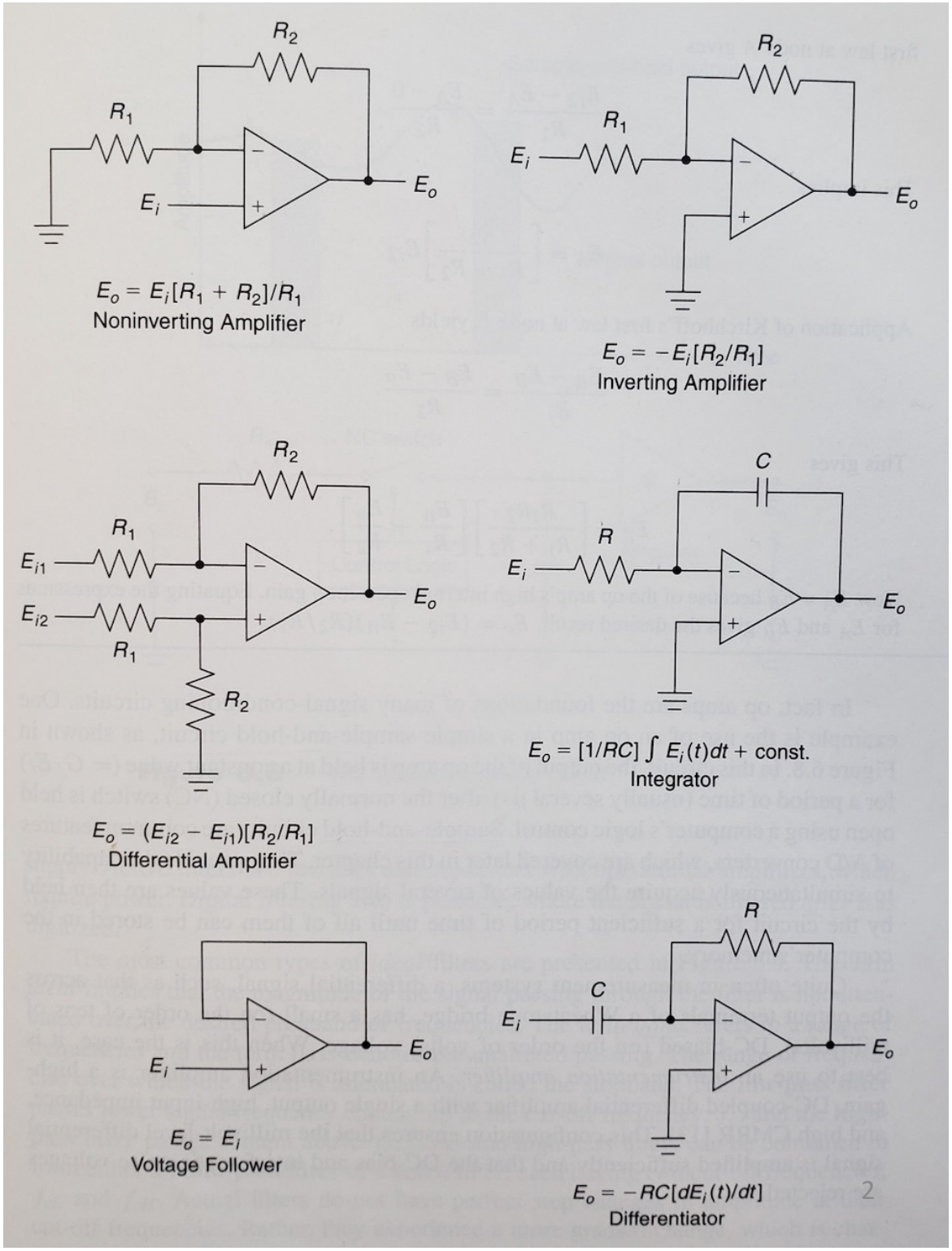
ENME 561/661

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Popular op-amp configurations

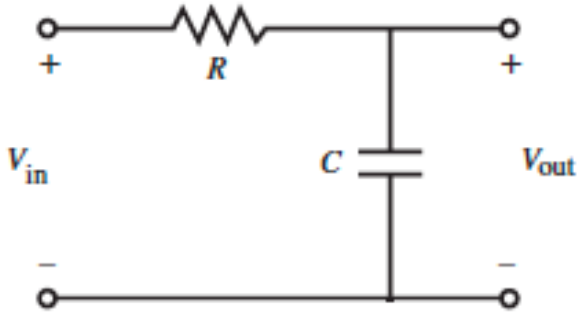


Noise: Unwanted component of a signal

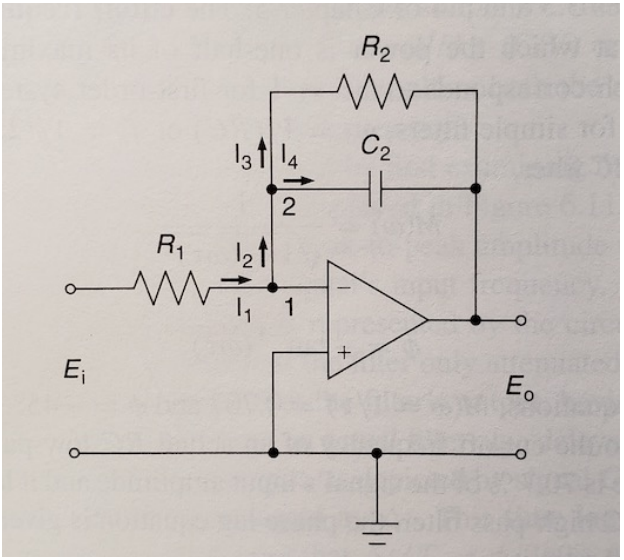
- Sources:
  - Sensor Channel noise
    - When measuring distance using IR sensors, reflections from the environment or background light can introduce noise.
    - When measuring acceleration, the motion of the object caused by building vibrations is considered noise.
  - Electric and magnetic field interferences
    - due to large voltage, current or magnetic sources in the vicinity of the signal
  - Noise inherent to the devices
    - For example, variations in the devices because of temperature

Analog active and analog passive filters

- Filters are used to remove unwanted frequencies
- Passive filters
  - Passive components:
    - resistors, capacitors, inductors
  - do not need an external source of power

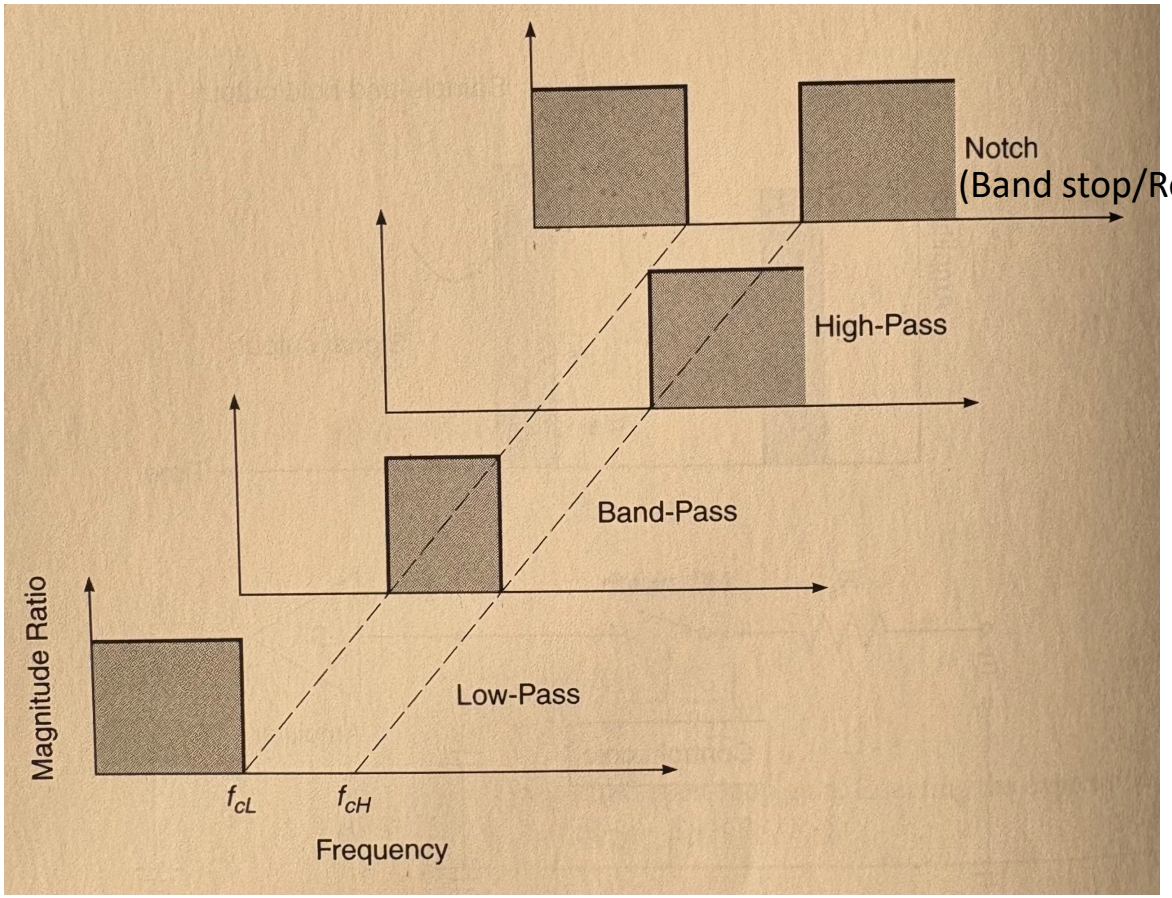


- Active filters
  - Passive and active components:
    - Operational amplifiers, resistors, capacitors
  - needs an external source of power



Ideal Filters

- *Band: range of frequencies*
- *Pass: unaltered passing*
- *Stopband: range of frequencies the signal is attenuated*



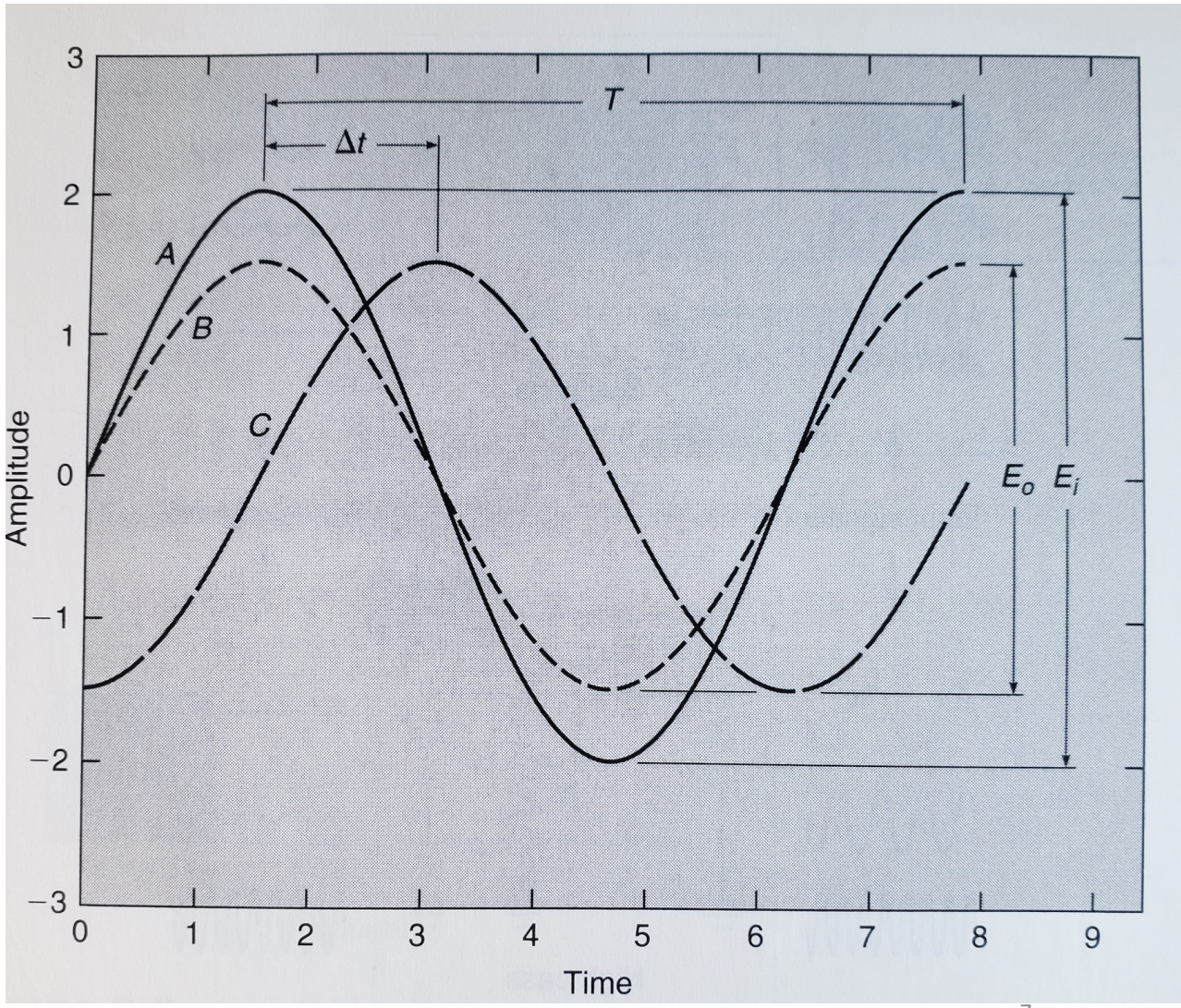
Cutoff frequency (half-power point)-A definition

- Frequency when the power of the output signal is attenuated to half of its input value
- The
- power of a sinusoidal signal is proportional to the square of the signal's amplitude

- At the cutoff frequency, The amplitude of the signal is attenuated by 29.3% (to 70.7% of its original value), about -3 dB:

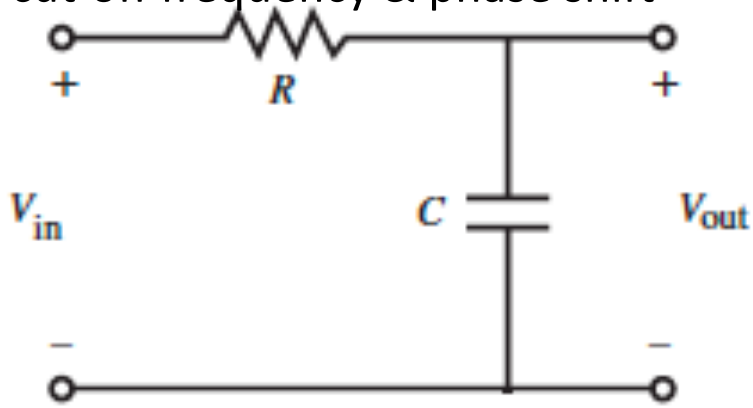
Filter input-output response characteristics

- A: signal
- B: Signal A with attenuated amplitude
- C: Signal A with attenuated amplitude and phase shift (filtered signal)



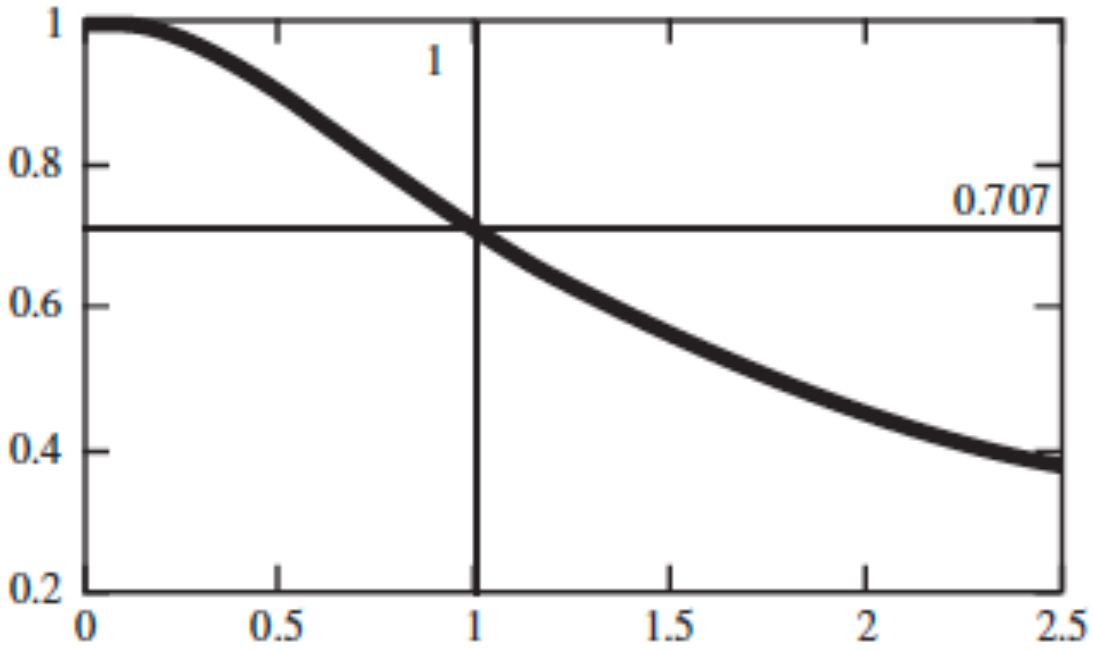
Analog Passive Low Pass Filter (Example: 1st order lowpass RC filter)

Derivation of the transfer function & amplitude ratio & cut-off frequency & phase shift



Amplitude ratio vs frequency ratio

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}$$

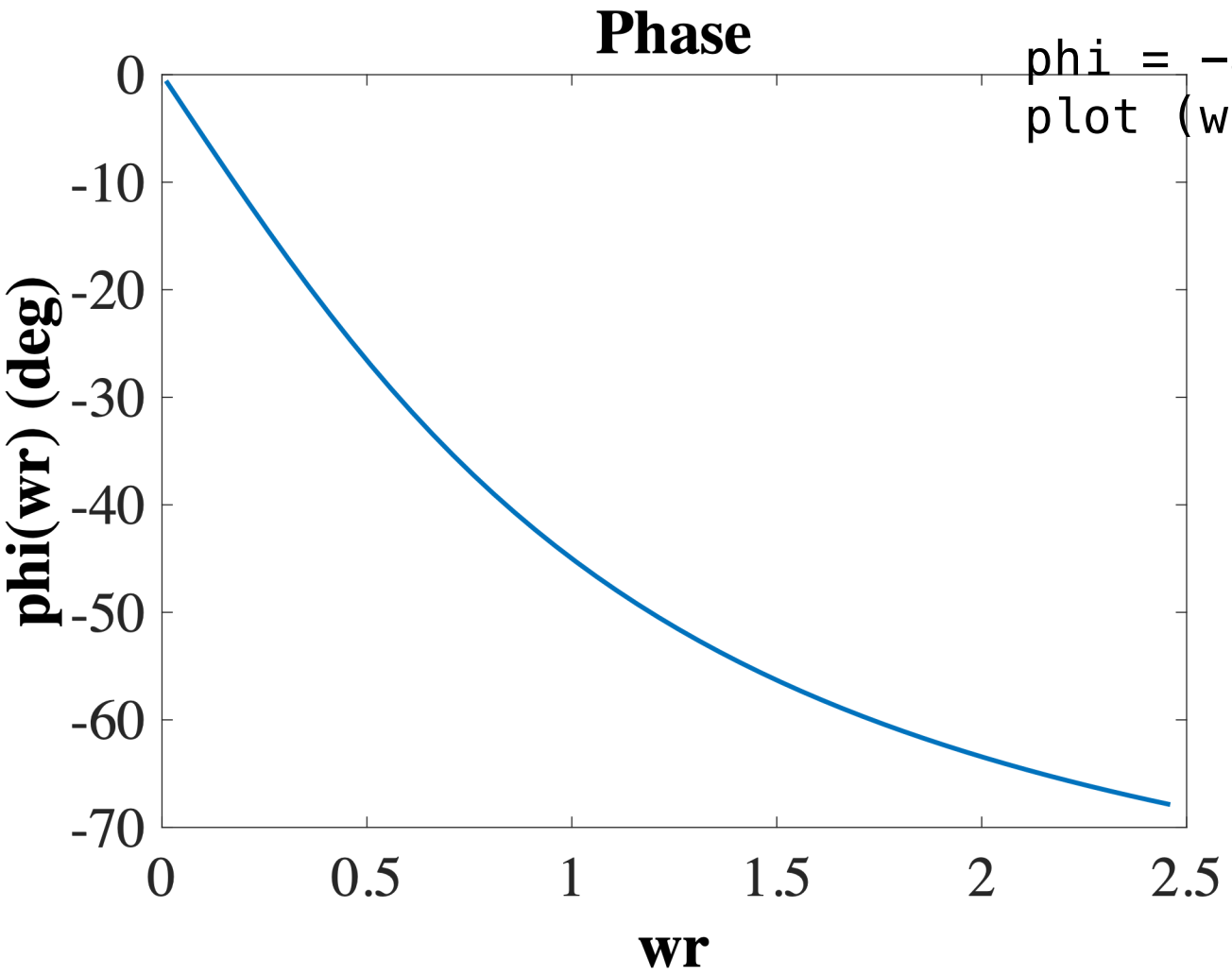


1st order Low pass filter

MATLAB plot: Phase lag

```
R = 10000;
C = 0.01 * 10^-6;
wc = 1 / (R*C);
wr = 0.01 : 0.05 : 2.5;
Ar = 1 ./ sqrt(1 + (wr).^2);
plot (wr, Ar)

phi = -atan(wr);
plot (wr, phi*180/pi)
```



Transfer Function of Filters (Examples)

- Low pass filter

$$\frac{V_{out}}{V_{in}} = \frac{1}{\tau s + 1}$$

- High pass filter

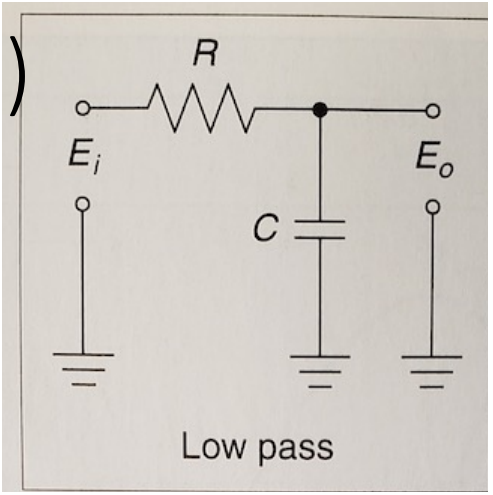
$$\frac{V_{out}}{V_{in}} = \frac{\tau s}{\tau s + 1}$$

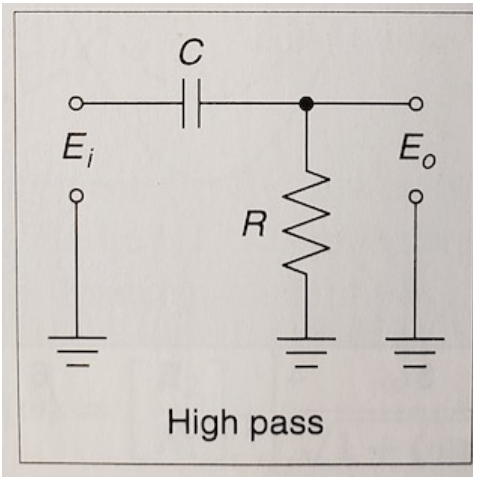
- Bandpass filter

$$\frac{V_{out}}{V_{in}} = \frac{\tau_1 s}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

- Bandstop (Notch) filter

$$\frac{V_{out}}{V_{in}} = \frac{\tau^2 s^2}{\tau^2 s^2 + 4\tau s + 1}$$





Reference

- Alexander, Charles K., Matthew NO Sadiku, and Matthew Sadiku. Fundamentals of electric circuits.
- Jouaneh, Musa. Fundamentals of mechatronics.
- Patrick F Dunn, Measurement and Data Analysis for Engineering and Science.
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- David G. Alciatore, Introduction to Mechatronics and Measurement Systems
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